

# Natural Language Processing

Fall 2026

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## Assignment 1

Tokenization, Preprocessing, and POS Tagging

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### General Instructions

- This is a **group assignment**. You must work in a group of **two or three students**.
- All the group members are expected to understand the complete implementation and analysis.
- Clearly mention the names and student IDs of all the group members in the report.
- Your code should be readable, appropriately documented, and reproducible.
- Follow the specific library restrictions given in each question.

### Question 1: Subword Tokenization from Scratch

You are provided with a small multilingual corpus [https://drive.google.com/drive/folders/1U-DfPULmLkIkp8TTbS32PYS5u0I0zJXG?usp=drive\\_link](https://drive.google.com/drive/folders/1U-DfPULmLkIkp8TTbS32PYS5u0I0zJXG?usp=drive_link) containing **English, Urdu, and Chinese text**, with the three languages mixed within the corpus.

Your task is to implement the following subword tokenization algorithms **from scratch**:

1. **Byte Pair Encoding (BPE)**
2. **WordPiece**

You may use standard Python libraries for basic text and file processing. However, you **must not use existing implementations of BPE or WordPiece** from libraries such as Hugging Face, SentencePiece, spaCy, NLTK, or similar packages.

### Hyperparameter Selection

You are responsible for selecting suitable values for the parameters and hyperparameters required by each tokenizer, including the **target vocabulary size  $K$**  and any other relevant design choices.

Your choices should take into account the characteristics of the provided corpus, including:

- the small corpus size;
- the presence of three different languages;
- differences among the English, Urdu, and Chinese writing systems;

- code-switching within the text;
- Unicode representation;
- the trade-off between vocabulary size and token fragmentation.

There is **no single correct value of  $K$** . You must justify your selected values and design choices in the report. You are encouraged to experiment with more than one value before deciding on the final configuration.

## Required Analysis

For both BPE and WordPiece, briefly discuss:

- the value of  $K$  and any other hyperparameters you selected and why;
- how changing the values of these hyperparameters affects the resulting vocabulary and learned subwords;
- any preprocessing or normalization performed before tokenization;
- whether the same preprocessing decisions are appropriate for English, Urdu, and Chinese;
- any interesting differences observed in the subwords learned for the three languages;
- differences between the vocabularies generated by BPE and WordPiece, including examples where the two algorithms learn different subword units.

## Output Files

Your program must generate the following two text files:

- `bpe_vocab.txt`
- `wordpiece_vocab.txt`

Each learned subword must appear on a **separate line**. The number of entries should correspond to the vocabulary configuration selected by your group.

## Submission for Question 1

Submit:

1. complete source code for BPE and WordPiece;
2. a report describing your implementation, experiments, hyperparameter choices, and observations. The report must be typeset in LaTeX using the ACM conference paper format. Submissions prepared in other formats will not be accepted. The report should clearly document your methodology, experimental choices, results, and analysis.;
3. `bpe_vocab.txt`;

#### 4. wordpiece\_vocab.txt.

**Important:** Marks will not be awarded simply for producing a vocabulary of the required size. Your implementation and report should demonstrate that you understand how tokenization decisions and hyperparameter choices affect the learned subwords, particularly in a multilingual corpus.

## Question 2: English Text Preprocessing, Frequency Analysis, and POS Tagging

For this question, use Jane Austen's *Emma* from the **NLTK Gutenberg Corpus**:

`austen-emma.txt`

You **may use NLP libraries such as NLTK** for this question.

You can access the corpus using:

```
import nltk
nltk.download('gutenberg')

from nltk.corpus import gutenberg
text = gutenberg.raw('austen-emma.txt')
```

### Part A: Text Preprocessing

Load the corpus and perform appropriate preprocessing. Your preprocessing should include:

- tokenization;
- conversion to lowercase;
- removal of punctuation and other non-word tokens;
- removal of English stopwords.

You may perform additional preprocessing if you consider it appropriate. Clearly mention and justify any additional preprocessing steps in your report.

Report the following **before and after preprocessing**:

- total number of tokens;
- total number of unique words.

### Part B: Word Frequency Analysis

Using the preprocessed corpus:

1. Report the **10 most frequent words** and their frequencies.
2. Report the **number of words that occur only once** in the corpus.
3. Briefly discuss any interesting observations about the most frequent words after stopwords removal.

Present the frequency results in a clear table.

## Part C: Part-of-Speech (POS) Tagging

Perform **POS tagging** on the text using an appropriate POS tagger available through an NLP library.

From the tagged text:

1. identify all words tagged as **nouns**;
2. create a list of the **unique nouns** appearing in the corpus;
3. report the **10 most frequent nouns** and their frequencies;
4. report the total number of noun occurrences;
5. report the total number of unique nouns.

You are responsible for determining which POS tags produced by your selected tagger correspond to nouns.

## Part D: Analysis

Briefly discuss:

- how preprocessing affected the vocabulary size and frequency distribution;
- whether stopword removal substantially changed the most frequent words;
- what types of nouns appear most frequently in the corpus;
- any preprocessing decision that affected the POS-tagging results.

Support your discussion with relevant examples from your output.

## Overall Submission

Each group must submit:

1. **Source Code** for both questions;
2. **Report** containing the required analysis, results, tables, and discussion for both questions;
3. `bpe_vocab.txt`;
4. `wordpiece_vocab.txt`.

**Group Size: Exactly two students.**

## Grading Rubric

### Question 1: BPE and WordPiece (60 Marks)

| Component                                                              | Marks     |
|------------------------------------------------------------------------|-----------|
| Correct implementation of BPE and WordPiece from scratch               | 25        |
| Hyperparameter selection and justification                             | 10        |
| Analysis of hyperparameter effects and comparison of BPE vs. WordPiece | 10        |
| Multilingual observations for English, Urdu, and Chinese               | 5         |
| Correct output files, code quality, organization, and reproducibility  | 10        |
| <b>Total</b>                                                           | <b>60</b> |

### Question 2: Preprocessing and POS Tagging (40 Marks)

| Component                                  | Marks     |
|--------------------------------------------|-----------|
| Correct preprocessing of the corpus        | 10        |
| Word frequency analysis                    | 8         |
| POS tagging and noun extraction            | 10        |
| Correct output/statistics and code quality | 8         |
| Analysis and presentation of results       | 4         |
| <b>Total</b>                               | <b>40</b> |

### Important Note

A badly formatted, poorly structured, or difficult-to-follow report will result in a deduction of up to **10 points** from the overall assignment grade.

**Overall Total: 100 Marks**